

GIZMO Operations Guide

Never lift the safety ground. This working guide is subject to SOP004, released grounding drawings, and the current work plan.

Why the metal structure must stay grounded

Keep the cryostat and beam structure connected to the required Building Ground during contractor work. Friction during material handling, moving belts, or dust flow can generate static charge. If the metal structure loses its ground connection, accumulated charge can discharge through a person or electronics, causing a shock or ESD damage.

A damaged powered tool or cable can also energize the structure. The approved safety-ground path carries fault current so electrical protection can operate. A connection adequate for draining static charge alone does not establish fault protection. GIZMO measures the grounding arrangement; it does not provide the safety ground.

Why repeat GIZMO tests

Construction changes the connections between Detector Ground and Building Ground. Tools, cables, scaffolding, or new metalwork can introduce unwanted conductive paths or change capacitive coupling. These changes can compromise detector noise performance. Regular tests find them while the recent work is still accessible and easier to correct.

The two checks serve different purposes: the bond check verifies the required safety connection; the GIZMO test checks isolation, shorts, capacitance, and coupling in the authorized test state. A high GIZMO impedance is not proof of a sound safety ground. The approved safety path, including the saturable-inductor assembly, must remain intact during isolation tests.

Contractor shift checks

At shift start and handover:

1. Inspect the grounding cable and connections for damage, looseness, corrosion, or obstructed contact. Have the designated trained checker verify the Building Ground bond using the approved meter method, test points, and acceptance limit.
2. Confirm the knife-switch position and tagout status against the work plan. Only authorized personnel may change them.
3. Record the check and arrange a clear test window with FNAL I&I/GIZMO. Remove tools and electrical cords that could bridge the grounds for the test.

Never use a meter's resistance mode on an energized circuit. A continuity beep alone is not acceptance of the safety bond.

FNAL I&I/GIZMO isolation tests

FNAL I&I/GIZMO owns readings, dashboard checks, and isolation tests. Agree and record recurring test windows before, after, or between shifts; repeat after work that could alter grounding or coupling.

1. Coordinate with the contractor. Establish and record the authorized knife-switch/tagout state while retaining the approved safety-ground path.
2. Allow the specified settling time. Verify fresh data, good measurement quality, calibration status, and excitation frequency.
3. Record resistance or HIGH Z, valid capacitance, coupling diagnostics, and alarm state. Compare with approved limits and the previous result at the same frequency and grounding state.
4. Investigate unexpected changes with the electrical lead. Restore and verify the required Building Ground connection, close out tagout, and record handback before contractor work resumes.

Log the time, checker, grounding state, readings, pass/fail result, action, handback, and next test in Procore Daily Log > GIZMO. HIGH Z is a range indication, not an exact resistance or automatic pass.

Faults, alarms, and missing measurements

- **Safety bond damaged, missing, or unverified:** stop affected work, keep clear, and notify the electrical lead. Resume only after defects are corrected, the bond is verified, and handback is authorized.

- **Unexpected GIZMO alarm or failed test:** preserve grounding, record the state and recent work, and notify FNAL I&I/GIZMO and the electrical lead. Investigate and retest under the work plan; stop affected work if safety is uncertain. Never remove a safety bond or change thresholds to clear an alarm.
- **Stale, uncertain, or missing data; missed test:** record the gap and arrange a valid local measurement. The facility must maintain instrument/network support and an approved local alternative to avoid delaying contractor work. If the required check cannot be completed, the responsible lead must resolve work authorization under the work plan; an old reading is not a pass.

The GIZMO beacon reports the configured impedance/phase alarm, not a diagnosis of an open safety ground. A dark beacon or running dashboard does not prove safe grounding.

Settings and connections

GIZMO is deployed under SOP004. Recorded settings: 1436 Hz excitation, runtime 0.5.1, OPC UA model 1.4.0; history starts July 27, 2026. Confirm the active settings before testing. The planned initial setup uses one saturable inductor.

After installation or maintenance, with the rack de-energized, verify labels and released drawings: Pearson CT to CT input on the intended safety-ground path; stimulus to Detector Ground and return to the designated reference; beacon/audio, network, and approved DC supply to their assigned connectors. The electrical lead verifies the inductor and safety-ground connections. Grounding, CT, stimulus, or calibration changes require the applicable work controls.

FNAL: server check

On dune-fd-test01, run:

```
/storage/gizmo-monitor/bin/status
```

Check upstream connectivity and the newest source timestamp and quality, as well as historian, dashboard, and disk status.

For a local dashboard connection:

```
ssh -N -L 127.0.0.1:18081:127.0.0.1:18081 dune-fd-test01
```

Then open <http://127.0.0.1:18081/> using the configured server access.

Historical status, September 30, 2026: remote measurements had been unavailable for 17 days after loss or damage of the mine network switch. The last retained sample was September 13 at 07:11:34 UTC, with Uncertain quality. This record does not establish today's condition.

FNAL: shutdown

Coordinate any loss of monitoring before shutdown. Use SOP004 and the deployed operating procedure to disable excitation and power down. Leave the permanent safety-ground path intact. Record the time and reason.

Complete before field use

The electrical lead and FNAL I&I/GIZMO must specify the bond-check method and limit, GIZMO test cadence, settling time, permitted limits, stale-data limit, local fallback method, support contact, recovery target, and handback authority in the work plan. These site values remain unassigned; this guide does not set them.

Safety background: [CCOHS: Static Electricity](#); [OSHA: Equipment Grounding](#). Site drawings and SOP004 govern the installation.